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TECHNICAL SPECIFICATION

REMOTE TERMINAL UNIT (RTU)
FOR TNB DISTRIBUTION NETWORK SUBSTATIONS

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1.0 PURPOSE

This specification defines the technical requirements for the Remote Terminal Unit (RTU), applicable to all denomination of RTUs or similar-class equipment to be installed in Distribution substations and SCADA-equipped assets in Tenaga Nasional Berhad (TNB).

It establishes the functional and performance requirements of the RTUs for monitoring and control of the physical devices in the power system in the TNB's Distribution Network substations.

Detailed project requirements will be specified in Bill Of Quantity (BQ) during the tendering stage.

2.0 GENERAL

2.1 PRIMARY SUBSTATIONS

Primary substations consists of Pencawang Masuk Utama, Pencawang Pembahagian Utama, Stesen Suis Utama, with Undergrounds and Overheads System with 110V DC station-supply system.

2.2 SECONDARY SUBSTATIONS

Secondary substations consist of Stesen Suis Utama, Pencawang Elektrik, with VCB or RMU switchgears with 30V DC station-supply system and Pole mounted assets, with 240V AC system.

2.3 Two types of DC Supply systems that are typically used for the essential systems including the power source for the RTU in the substations are:-

i. 110V DC Center-Tap System, typically found in the Primary substations, with nominal voltage of 125Vdc.
ii. 30V DC Center-Tap System, typically found in the Secondary substations, with nominal voltage of 35Vdc.

Other assets operating in other AC or DC voltages will be specified in the tender documents, when necessary.

2.4 SCADA SIGNAL LIST

The Distribution SCADA signal list is subject to standard signal name and site requirement during project implementation. Successful tenderer shall use these standard signal names in the RTU database configurations during project implementation.

2.5 CONFORMANCE

Proof of Conformance to all Standards specified herein and all type test results shall be submitted during Product Certification, Product Inspection processes and tendering processes.

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3.0 DESIGN & CONSTRUCTION

The RTU shall be of modular design, when the Input/Output (IO) sizing requires it, of low power consumption and low heat generation, and suited for electric power distribution substation. Failure of any IO module shall not affect the performance of other modules.

The RTU, when required in the tender, shall be installed in a specific RTU panel described later in this specification.

Integrated RTU design in a small form-factor, may be offered when small IO count (< 16) are required. This design requires a robust construction to be installed in it's own panel or in other equipment panels.

Some deviations or exclusions of physical attributes and functional requirements stated herewith may be required depending on type and operational requirements which will be specified in the tendering stage. Examples of these are the type and quantity of Communication ports, type of form factors, type of RTU Power Supply System and minimum Input and Output (IO) quantity.

4.0 SAFETY REQUIREMENTS

The RTU shall conform to the requirements of either:-

i. IEC 61010-01 Safety requirements for electrical equipment for measurement, control and laboratory use.

ii. IEC 60950-1 (2005-12 or latest ed.) Information Technology Equipment – Safety – Part 1: General Requirements.

5.0 CYBER SECURITY REQUIREMENTS

The RTU shall be IEC 62351/ ISO/IEC27001/ NERC-CIP ready in order to comply with the cyber security objectives and requirements, which includes the support for secure SCADA protocols.

6.0 RTU ENVIRONMENTAL REQUIREMENTS

6.1 General Environmental Requirement

RTU modules and enclosures used shall be suitable for the climate and environmental conditions pertaining to Malaysia and shall be:-

- Vermin-proof
- Of tropical design
- Of robust construction
- Insensitive to changes of temperature, humidity, vibration etc.

The following sections describe the type testing requirements to be met by the RTU.

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6.1.1 Atmospheric Environment

6.1.1.1 Operating Temperature

The RTU shall be able to operate normally within the operating ambient temperature 0 to +70 °C, with no reduction in reliability, without the aid of any forced cooling systems.

The RTU shall be tested for Operating Temperature requirement to the standards.

Requirement	Method	Criteria	Following Test Standard
Temperature	Dry Heat	Up to (70 ± 2) °C at 96 hours	IEC 60068-2-2

6.1.1.2 Relative Humidity

The RTU is required to operate satisfactorily over the relative humidity criteria and tested to the standards. Maximum absolute humidity 28 g/m^3.

Requirement	Method	Criteria	Following Test Standard
Relative Humidity	Damp Heat, Steady State	Severity class (40 ± 2) °C, (93 ± 3) % RH, 10 days.	IEC 60068-2-78
Relative Humidity	Damp Heat, Cyclic	12 + 12 hour cycles between 25 to 55 °C at 95% RH, 6 cycles.	IEC 60068-2-30

6.1.2 Mechanical Environment

The RTU shall withstand the mechanical test. The test shall be performed on complete assemblies and during the tests, performance checks shall be made. Any resonance detected that are likely to be detrimental to performance or equipment life shall be rectified.

Requirement	**Method**	**Criteria**	**Following Test Standard**
Vibration Response and Endurance Test	Vibration (sinusoidal)	Severity Class 1, frequency range 10 to 150Hz with a severity of not less than 4.9m/s sq. at nominal sweep rate of 1 octave/min	IEC 60068-2-6 IEC 60255-21-1
Shock Response, Withstand Test, Bump Test	Shock	Severity Class 1	IEC 60068-2-27 IEC 60255-21-2

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6.1.3 Electrical Environment

The RTU shall be designed for safe operation in high voltage environment. Data communication ports shall be tested to withstand disturbance test without permanent corruption of data, and subsequent delay of data transfer.

The requirements in this section are based on guidelines established by IEC 60870-2-1 and IEC TS 61000-6-5 Electromagnetic Compatibility (EMC) Part 6-5: Generic Standards – Immunity for power station and substation environments.

The following type test requirements shall apply to all modules covered under the specification's scope of supply unless specifically stated otherwise.

Tests shall be performed on the RTU system (which means a complete functional RTU with all the processor, serial communications and input/output modules assembled) if the test specifically mentions it.

During the disturbance, the following performance criteria shall be met:-

i) Analogue Inputs - variations shall not exceed twice the class index for inputs associated with analogue quantities.
ii) Digital type inputs - no variation in indicated state should occur.
iii) Pulse count inputs - no spurious count and no incorrect count value reported.
iv) Control outputs - no unintentional change in output or reported status.
v) Analogue outputs – no unintentional change in analogue output value.
vi) Communications link – retries are permissible but no erroneous information shall be exchanged.
vii) RTU operation – continuous operation with no stops or resets.

After the test has been applied, the equipment under test must recover and return to full operational status.

Requirement	Method	Criteria	Following Test Standard
Non-Damage (Insulation)	Performed on de-energized equipment	For all modules as mentioned above	IEC 61000-6-5 IEC 60780-2-1 (Immunity)

6.1.3.1 Insulation (Non –Damage)

6.1.3.1.1 Dielectric

The test voltages shall be applied in the following manner:-

- Between each isolated input and earth.
- Between each isolated input and all other inputs connected together.
- Between each isolated output and earth.
- Between each isolated output and all other outputs connected together.

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- Between all isolated inputs connected together and all isolated outputs connected together.
- Between power supply terminals and all isolated input and output terminals connected to earth

Requirement	Method	Criteria	Following Test Standard
Non-Damage (Insulation) Dielectric Test Voltage	For all ports/terminals as mentioned above	• Insulation resistance to earth: 100 Mohm at 500 V dc • Withstand 2kV ac to earth for 1 minute • Withstand 1kV ac across open relay contact circuits.	IEC 60255-27:2013 (Immunity)

6.1.3.1.2 **Impulse Voltage**

The test voltages shall be applied in the following manner:-

- Between each terminal and the safety earth (Common mode)
- Between individual terminals of all independent circuit (including power supplies) (Common mode between separate circuits)
- Between signal terminals of the same circuit (Differential mode)
- Between all combinations of DC power supply terminals and signal terminals for I/O of greater than 24 V (External supply) (Differential mode)

Requirement	Method	Criteria	Following Test Standard
Non-Damage (Insulation) Impulse Voltage	For all ports/terminals as mentioned above	1.2/50microsec waveform, 0.5 joule: • Test voltage of 5kV for common mode test. • Test voltage of 1kV for differential mode test	IEC 60255-22-5 (Immunity)

6.1.3.2 **Electromagnetic Compatibility (Non-maloperation)**

Tests shall be performed on the RTU system (which means a complete functional RTU with all the processor, serial communications and input/output modules assembled) if the test specifically mentions it.

During the disturbance, the following performance criteria shall be met:-

i) Analogue Inputs - variations shall not exceed twice the class index for inputs associated with analogue quantities.
ii) Digital type inputs - no variation in indicated state should occur.
iii) Pulse count inputs - no spurious count and no incorrect count value reported.
iv) Control outputs - no unintentional change in output or reported status.
v) Analogue outputs – no unintentional change in analogue output value.

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vi) Communications link – retries are permissible but no erroneous information shall be exchanged.
vii) RTU operation – continuous operation with no stops or resets.

After the test has been applied, the equipment under test must recover and return to full operational status.

Requirement	Method	Criteria	Following Test Standard
Non-Maloperation (Electromagnetic Compatibility)	Performed on energized equipment	For all modules as mentioned above	IEC 61000-6-5 IEC 60780-2-1 (Immunity)

6.1.3.2.1 **Mains Frequency Voltage**

The test voltages shall be applied in the following manner:-

- Between each terminal and the safety earth (Common mode).
- Between individual terminals of all independent circuit (including power supplies) (Common mode between separate circuits).
- Between signal terminals of the same circuit (Differential mode).
- Between all combinations of DC power supply terminals and signal terminals for I/O of greater than 24 V (External supply) (Differential mode).

Requirement	Method	Criteria	Following Test Standard
Mains Frequency Voltage	For all ports/terminals as mentioned above	Level 4	IEC 61000-4-16 (Immunity)

6.1.3.2.2 **High Frequency Disturbance (Damped Oscillatory Waves)**

The test waveform is a 1 MHz damped oscillatory waveform from a 200 ohm source. The decay envelope shall be down to 50% in 3 to 6 cycles and the repetition rate shall be 400 Hz.

The test voltages shall be applied in the following manner:-

a) For Common Mode:
   - Between each terminal and the safety earth
   - Between individual terminals of all independent circuit (including power supplies) (Common mode between separate circuits)
   - Test voltage of 2.5kV for 2 seconds

b) For Differential Mode:
   - Between signal terminals of the same circuit
   - Between all combinations of DC power supply terminals and signal terminals for I/O of greater than 24 V (External supply)
   - Test voltage of 1kV for 2 seconds

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Requirement	Method	Criteria	Following Test Standard
High Frequency Disturbance (Damped Oscillatory Waves)	For all ports/terminals as mentioned above.	Level 3	IEC 61000-4-12 or IEC 60255-26 (Immunity)

6.1.3.2.3 **Electrostatic Discharges**

The equipment shall meet the requirements as mention below:

Requirement	Method	Criteria	Following Test Standard
Electrostatic Discharges (ESD)	• Test voltage of 6kV for contact discharge test • Test voltage of 8kV for air discharge test	Level 3	IEC 61000-4-2 (Immunity)

6.1.3.2.4 **Radiated Electromagnetic Field Disturbance (non-maloperation)**

The equipment shall meet the requirements as below:

Requirement	Method	Criteria	Following Test Standard
Radiated Electromagnetic Field Disturbance	Tests at a field strength of 10 V/m over the frequency range of 80 to 3000 MHz (80% AM; 1 kHz)	Level 3	IEC 61000-4-3 (Immunity)

6.1.3.2.5 **Fast Transient**

The equipment shall meet the requirements as below:

Requirement	Method	Criteria	Following Test Standard
Fast Transient	Burst: Full test value of 4kV shall be applied for the power supply ports and signal ports (analogue input/output, digital input/output, communication ports)	Level 4	IEC 61000-4-4 (Immunity)

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6.1.3.2.6 **Conducted disturbances induced by radio frequency fields, amplitude modulated**

The equipment shall meet the requirements as below:

Requirement	Method	Criteria	Following Test Standard
Conducted disturbances induced by radio frequency fields, amplitude modulated	10V; 150kHz to 80 MHz; 80% AM; 1 kHz.	Level 3	IEC 61000-4-6 (Immunity)

6.1.3.2.7 **Power frequency magnetic field**

The equipment shall meet the requirements as below:

Requirement	Method	Criteria	Following Test Standard
Power frequency magnetic field	30 A/m continuous; 300 A/m for 3 s pulse.	Level 4	IEC 61000-4-8 (Immunity)

6.1.3.2.8 **Surge immunity**

The equipment shall meet the requirements as below:

Requirement	Method	Criteria	Following Test Standard
Surge Immunity	100 A/m continuous; 1000 A/m for 1 second, for both signal and power ports.	Level 4	IEC 61000-4-5 (Immunity)

6.1.3.2.9 **Conducted interference voltage, aux. Voltage**

The equipment shall meet the requirements as below:

Requirement	Method	Criteria	Following Test Standard
Conducted Interference Voltage, aux. Voltage	30 MHz to 1000 MHz	Class A	CISPR 32 (Emission)

6.1.3.2.10 **Interference field strength**

The equipment shall meet the requirements as below:

Requirement	Method	Criteria	Following Test Standard
Interference Field Strength	1000 MHz to 6000 MHz	Class A	CISPR 32 (Emission)

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6.1.3.2.11 **DC supply interruptions IEC 61000-4-29**

The equipment shall meet the requirements as below:

Requirement	Method	Criteria	Following Test Standard
DC Supply interruptions	• An interruption of 50 ms (100% u) • A voltage dip of 30% for 0.1s • A voltage dip of 60% for 0.1 s	Class C	IEC 61000-4-29 (Immunity) Or IEC 60255-26

The equipment shall withstand a 20 ms interruption in the auxiliary supply under normal operating conditions without de-energizing. The 20ms interruption shall be repeated three times with a maximum interval of 60 ms in between interruptions.

6.1.3.2.12 **AC ripple on DC supply IEC 61000-4-17**

The equipment shall meet the requirements as below:

Requirement	Method	Criteria	Following Test Standard
AC Ripple on DC supply	15% ripple on the DC auxiliary supply at 50Hz	Class A	IEC 61000-4-17

7.0 RTU PERFORMANCE REQUIREMENTS

7.1 The RTU shall be manufactured to the highest possible quality so as to achieve a minimum of 10 years normal useful life.

7.2 All basic critical RTU functionalities specified in this document shall be intrinsic embedded RTU functions that will be made available as in-built functionality without using logic programming or external algorithms.

7.3 RTU shall be designed and manufactured in a manner as to enable standard operation and maintenance tasks to be carried out effectively and safely.

7.4 The availability of the RTUs shall be ≥ 99.95% (IEC 60870-4 Table 2 Class A3) over the normal useful life expectancy of the RTU (not less than 10 years) exclusive of communication channel and other planned down time.

7.5 Mean Time Before Failure (MTBF) and the Mean Time To Repair (MTTR) figures used to establish the RTU availability shall be declared.

7.6 A known standard for the components reliability such as MIL-HDBK-217 shall be used to determine the MTBF and Failure Rate (α) for each module that constitutes the RTU. MTBF and Failure Rate data derived from field reliability

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studies shall only be accepted if the module has already been in commercial service for at least 3 years.

7.7 For the purpose of calculating the MTBF for a complete RTU system, the failure of any module within the RTU shall constitute the failure of the whole RTU.

7.8 The RTU Vendor shall declare the configuration of the RTU that shall meet the reliability requirements as defined in this specification.

8.0 RTU POWER SUPPLY

8.1 The power supply to run the RTU shall be provided by TNB via the substation's Power Supply system as mentioned in Para 2.3, *unless stated otherwise in the tendering stage.*

8.2 The power supply shall be connected to the RTU panel with separate suitably rated MCBs for

i. The operation of the RTU and isolated power for the "wetting" of contact inputs.
ii. The operation of the coils of the plant-side heavy duty interposing relays of the control output points (Refer Control Output Subsystem Requirements).
iii. The operation of communication equipment such as modems and routers.

8.3 The RTU power supply module shall be able to operate from the substation power supply with a tolerance of -15% to +20% of nominal input voltage and maintain full normal operation over the input supply range.

8.4 When the RTU experiences a fault within its power supply system, it shall contain the fault within the RTU and not cause disruption to the Station's DC supply such as causing a short circuit on the supply or earth faults.

8.5 The RTU shall recover from and resume normal operation in the case of it experiencing a momentary (≥ 20ms) disruption of the power supply input. The recovery shall be automatic without requiring manual intervention.

8.6 The RTU power supply system shall have adequate protection to prevent overloading on the input supply.

8.7 Suitably rated input transient overvoltage protection shall be incorporated into the RTU's power supply design.

8.8 The RTU shall sustain normal operation during slow variations of the power supply voltage from and to the nominal value to it's design cut-off voltage. Below the cut-off voltage, the RTU will shut down in an orderly manner without generating any spurious alarms, wild fluctuations in analog readings as well as unintended control operations.

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8.9 Power supply bus in RTU chassis shall be carefully routed and each bus shall be covered. It shall not be possible to inadvertently short busbars either between themselves or to earth.

9.0 RTU CLOCK

9.1 The RTU shall be equipped with a real time clock system, with full calendar (including leap year support), to allow the RTU to: -

i) Accurately provide the correct date and time for its operation.
ii) Accurately time-tag data acquisition and control events such as digital input changes, control executions and analog readings.
iii) Start up with the current date and time without the need for any manual settings or adjustments.
iv) Schedule execution of internal programs and algorithms.
v) Accurately record system events such as program start or end time, system faults etc.
vi) Maintain time synchronization with the SCADA Master System to be within 10ms for accurate and correct interpretation of time-tagged events.

9.2 The RTU clock resolution (the minimum discernable change in time generated) shall be ≤ 1 msec (IEC 60870-4 Table 7 Class TR4).

9.3 The clocks shall have an accuracy of ±2 ppm and shall not drift more than 7.2 msec per hour.

9.4 The RTU shall be able to accept both GPS time synchronization and a SCADA Master System time synchronization.

9.5 The RTU support of time-synchronization is to enable automatic and continuous updating of the RTU's calendar date and time in the following format: year/ month/ date/ hour/ minute/ second /millisecond.

9.6 For GPS time-synchronization using external, standalone, commercial off-the–shelf (COTS) GPS receiver clock, the RTU shall accept a commercially available non-proprietary Time Code synchronization.

9.7 For SCADA time-synchronization using the SCADA Master System, the RTU shall accept all time-synchronization command functions available in the protocol, including the Delay Acquisition/Measurement function in Individual or Broadcast addressing type.

9.8 The RTU shall be equipped with time synchronization arbitration facility such that it shall be possible to assign any synchronizing source (GPS or Master Stations) as the primary time synchronizing source. Upon failure of the primary source, the next ranked source's time shall be used for time synchronization. The failure shall operate a Primary Source Synchronization Lost Alarm. On recovery of the primary source the alarm shall be automatically reset. The primary source for time synchronization for RTUs equipped with a GPS receiver shall be the GPS time source.

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9.9 The RTU shall has a Time Synchronization Error function. The Time Synchronization Error is the maximum difference between the time used for synchronization, i.e. the time of the synchronizing clock (GPS or SCADA Master), and the time of the clock that is being synchronized. The maximum allowable time synchronization error with GPS time shall be less than one millisecond, and with SCADA Master is one millisecond. For RTUs with I/O modules that have their own clocks, the maximum error between its clock and the synchronizing clock shall be less than one millisecond.

9.10 The RTU shall monitor it's clock and shall tag it's time element invalid if it's clock drifted more than 7.2msec per hour since it is last synced. Persistent drifting of clock shall operate a Time Sync Error Alarm and shall be reported as a virtual DI point alarm.

9.11 Summer time clock adjustment in the RTU shall be disabled by default.

10.0 COMMUNICATION REQUIREMENTS

The RTU shall be capable of communicating as a Controlled Station or a slave, to the Distribution Network Control Centre (NCC) SCADA Master Station using IEC 60870-5-104, IEC 60870-5-101 and DNP3 (Level 3) SCADA Protocols Standards without any external protocol converter. Interoperability or Device Profile documents shall be submitted during tendering and shall be finalised in the Statement of Work process phase for the successful tenderer.

The RTU shall be capable of communicating as a Submaster, to Satellite RTUs or IEDs using IEC 60870 and DNP3 protocols as well, independent of the protocol to the SCADA Master.

The selection of communication protocol or protocols for the RTU with the SCADA Master System or the RTU as a Submaster and the quantity of communication ports, shall be specified during project stage on per project basis.

10.1 Communication Ports

10.1.1 The RTU shall support at least two Ethernet 10BaseT ports which may be configured and used independently to support communication with multiple SCADA Master Station, IEDs, Satellite RTUs, Configuration Tools and Maintenance Tools.

10.1.1.1 The Ethernet ports shall support multi TCP/UDP ports for various applications running at once.

10.1.1.2 All TCP/UDP ports shall be configurable to be OPEN for running applications or CLOSE for security purposes.

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10.1.2 The RTU shall support at least two RS-232C serial ports which may be configured and used independently to support communication with multiple SCADA Master Station and Satellite RTUs.

10.1.3 The RTU shall support internal or external cellular modem, for Remote Configuration and Maintenance purposes, with at least one cellular connectivity, via normal SIM card or programmable embedded SIM chip, *unless stated otherwise in the tendering stage.* Multi SIM connectivity is preferred.

10.1.4 The RTU shall be configured to reply to whichever port that it is polled from.

10.1.5 The Ethernet port and the serial port interfaces must be physically designed to be different from one another to prevent wrongful insertion of cable plugs.

10.1.6 All communication ports shall be able to be configured as main and standby or as redundant ports.

10.1.7 When the communication ports are configured as Main and Standby ports, the RTU shall monitor communications with the Master Station and in the event of a loss of communications on the main channel, failover from the main communication channel to a standby channel shall occur. Configuration parameters such as port selection and timeout period shall be user configurable.

10.1.8 Assignment of communications protocols to individual communication ports shall be done using the RTU Maintenance or Configuration Tools.

10.1.9 Each communication port shall function independently from other ports. Each port shall be unaffected by simultaneous activity on other communication ports.

10.1.10 The RTU shall support simultaneous communication with multiple master stations using various multi-port schemes, common or different communication protocols and at different baud rates.

10.1.11 The RTU shall be designed such that when configured for communication with multiple SCADA Master Stations, the master stations can share a single RTU point independently and that the operation of one shall not affect the others. The configured IO databases shall also be available on all communication ports.

10.1.12 Communications error to any one master station shall not affect communication to the other master station.

10.1.13 Visual indications of transmit and receive activity for all serial communication ports shall be provided.

10.1.14 All wired communication ports shall be equipped with suitably rated surge protectors.

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10.2 RTU ALARM DETECTION

10.2.1 The RTU shall include a continuous extensive self-testing.

10.2.2 The RTU shall detect the malfunction of the RTU such as (but not limited to) module failures, main communication failure, Satellite RTU or IED communication failure, and buffer failure.

10.2.3 These malfunctions or failures shall be indicated by visual alarms and as virtual points that can be mapped to the RTU IO points. In the event of a failure, a common visual Summary Alarm and a virtual IO alarm should be activated. On clearing the fault, the alarm should automatically be deactivated and reset.

10.2.4 Each alarm shall be indicated by visual display on the faulty modules or cards and to be reported in detail in the Maintenance Tools software as well.

10.2.5 Two auxiliary contacts shall be provided to indicate an RTU FAULT condition to external visual connections, for example to LEDs/lamps on the outside of the RTU panel.

11.0 DIGITAL INPUT SUBSYSTEM

11.1 The Digital Input Subsystem shall be made up of modules and termination assemblies to monitor the state of devices and alarms through the sensing of the state of contacts. These contacts from the plant equipment are voltage free contacts.

11.2 Each digital input shall function independently of any other and may be configured as any of the following type of digital point: –

- Single bit digital input
- Double bit digital input

11.3 Not more than 8 digital input points shall share one signal common.

11.4 The digital inputs shall be wired to voltage-free contacts from the plant. The RTU shall provide the wetting voltage to these contacts.

11.4.1 The RTU shall use appropriate protection devices to provide the sensing voltage for the digital inputs such that faults from the plant wires, such as earth faults and incorrect wiring resulting in an introduction of external voltages into the RTU, will not affect the supply to other modules or damage to any of the RTU modules.

11.5 All digital input circuits shall be individually optically isolated.

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11.6 Each input shall be monitored by an LED that indicates when the monitored input contact is closed.

11.7 It shall be possible to invert digital inputs, on a per point basis, so that a "B" or normally closed contact can be reported as an "A" or normally open contact.

11.8 All digital inputs shall be scanned by the RTU at least once every millisecond.

11.9 The RTU must be able to detect and process any event of at least 5ms in duration.

11.10 The RTU shall have a programmable debounce filter that can be configured to reject pulses from 5ms to 25 ms in width, in 1ms steps. A digital filter is preferred for this application.

11.11 It shall be possible to manually disable reporting of digital input points via configuration on a single and double point input basis.

11.12 The RTU shall be equipped with anti-chatter algorithm to reduce nuisance reporting of noisy or chattering plant contacts. The relay chatter filter will have two configurable parameters; the number of changes, *Nc* and the chatter filter period, *Tc,* to define its filtering function. If a point changes state more than *Nc* times within a period of *Tc*, the system will mark the point as chattering and disable it so that no further changes of state or events will be recorded until the point recovers from chattering. The relay chatter filter must be able to automatically restore monitoring of the point when it detects that it is no longer in a chattering state.

11.13 The *Double Bit Processing Time* is the time interval the detecting system waits, when a status change occurs for a double-bit point to ensure that double bit transition or intermediate states that occur during the interval is not reported. The transition states are "00" and "11". Valid states are "01" and "10". When a double-bit point changes state, the detecting system waits for a time interval equal to the double-bit processing time. If the point changes state to a transition state and then to a final valid state during that interval, the digital input subsystem will not report the transition state. If the point fails to reach a valid state at the end of the interval, the subsystem will report the transition state as the final state.

11.14 The Double Bit Processing time shall be a configurable parameter ranging from 0 to 255 seconds (default 10 seconds).

11.15 All digital input points, single-bit or double-bit points, shall be equipped to provide sequence-of-events time tagging in addition to the normal function of the DI points. It shall be possible to assign any DI data points in the RTU as an SOE point.

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11.16 When a change of state occurs, the event shall be identified with a point identification number, the type of event, and the time of occurrence in hours, minutes, second, and milliseconds.

11.17 In order to correctly interpret time-tagged events, it shall be possible to identify events that have been time-tagged when the RTU has lost time synchronization or when the RTU's clock has exceeded the specified drift.

11.18 All events must be time-tagged with the time of occurrence and shall be reported with time-tagged irrespective of the Data Class they are assigned to.

11.19 The maximum variation between the time that the system tags an event and the actual time the event happens is defined as the "Event Time Tagging Error". It is normally dependent on the frequency at which the digital input point is scanned as well as the clock resolution. The Event Time Tagging Error should not be more than one millisecond.

11.20 Separating Capability is the minimum time by which events must be separated such that the sequence of their occurrence can be determined correctly. The time separating capability shall be ≤ 5 milliseconds for events occurring within a substation (IEC 60870-4 Table 6 Class SP3).

11.21 SOE Time Resolution is the minimum time by which two events must be separated in order that the corresponding time tags are different. The Time Resolution shall not exceed one millisecond.

11.22 The RTU shall have a Main SOE buffer located in the RTU's main processor or similar module. This Main SOE buffer shall contain time-tagged sequential memory to store SOE events from all the digital input modules and subsystems within the RTU. The RTU shall use this Main SOE buffer to report SOE data to the Host. Event shall be stored in the chronological order in which they occur. This cyclic memory buffer shall be separate from all other RTU data buffers. When the cyclic buffer becomes full of entries that have not been reported to the Host the event archiving ceases and the RTU notes it is now 'losing' events. An alarm indication shall be provided to indicate that the RTU SOE data buffer has overflowed. Events that have been reported to the Host can be overwritten by new events.

11.23 The size of the Main SOE buffer shall be configurable and shall have enough memory to support archiving of at least ten times the number of installed digital input points.

11.24 For distributed I/O modules, digital input modules with individual processors as well as RTUs that employ distributed processing, individual SOE buffers to collect and store SOE data from its own digital input subsystem before being transferred to the Main SOE buffer must be provided.

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11.25 The Digital Input Subsystem shall be designed to facilitate the replacement of digital input modules without necessitating the re-termination and re-testing of field cabling.

11.26 Suitable terminal blocks with isolation and test points complete with cabling, shall be used for termination of field I/O cablings and shall be able to accommodate up to 2.5 mm sq. solid or stranded wire including spare points. The electrical ratings shall be 15 Amps/300 Volts and dielectric withstand shall be 3000 Volts.

12.0 BINARY CODED DECIMAL (BCD)

12.1 The RTU shall support Binary Coded Decimal inputs through the sensing of the state of contacts.

12.2 The BCD functionality shall be integrated into the Digital Input module.

12.3 Inputs to the BCD will be dry contacts. The contact sensing voltage options shall be the same as for digital inputs previously specified.

12.4 The number of BCD digits shall be configurable from one decade (0...9) to four decade (0000...9999) without sign.

12.5 A settling time shall be configurable for each BCD input for validating a stable BCD input indication. The settling time range configurable for the BCD shall accommodate the typical tap-changing operation times of the transformers in TNB's power system. Any change on an input channel shall retrigger the settling time.

12.6 Suitable terminal blocks with isolation and test points complete with cabling, shall be used for termination of field I/O cablings and shall be able to accommodate up to 2.5 mm sq. solid or stranded wire including spare points. The electrical ratings shall be 15 Amps/300 Volts and dielectric withstand shall be 3000 Volts.

13.0 COUNTERS/ACCUMULATORS

13.1 Pulse counters shall be provided for accumulating kWh and KVArh readings or as trip/operation counters for circuit breakers. Digital pulses from Purchaser-furnished meter contacts (Form A or C) shall be processed to digitally store the accumulated pulses.

13.2 The counter functionality shall be integrated into the Digital Input module.

13.3 It shall be possible to configure the pulse counter inputs as:

- Form A inputs
- Form C inputs

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- Incrementing on each pulse cycle (count on 0 to 1 transitions only)
- Incrementing on each pulse half cycle (count on 0 to 1 and 1 to 0 transitions)

13.4 A Form C counter input uses two sets of contacts; a normally open contact (Form A) and a normally closed contact (Form B) and shall record a count when both inputs change states. A Form C counter input shall not register a count when only one of the contacts changes state.

13.5 The module shall be provided with the option and the capability to retain counter values when RTU is switched off.

13.6 Inputs to the counters will be dry contacts. The contact wetting voltage options shall be the same as for digital inputs previously specified.

13.7 The maximum input pulse rate will be at least 20 pulses per second.

13.8 Each pulse counter readings shall be maintained in 32-bit registers.

13.9 An automatic 'roll-over' capability is required.

13.10 Each counter within the RTU shall be able to accumulate any input occurring while the counter is being read, so that no input counts will be lost.

13.11 The RTU shall be capable of responding to FREEZE commands generated internally within the RTU at a regular time interval. The FREEZE command shall cause the contents of the counter register to be copied to a separate freeze register. The system shall have the capability to read the freeze register and transmit the information to the master station at any time without changing the contents of the register.

13.12 The RTU shall be capable of responding to RESET commands generated internally within the RTU at a regular time. The RESET command effectively zeroes the counter.

13.13 The RTU shall be capable of responding to FREEZE & RESET commands generated internally within the RTU at a regular interval. A reset of the count shall take place once it has been frozen and reported. The time stamp at the time of freeze shall be recorded.

13.14 The time period shall be user configurable in minutes on a channel-by- channel basis, in the range 1 – 255 minutes (default 30 minutes).

13.15 Suitable terminal blocks with isolation and test points complete with cabling, shall be used for termination of field I/O cablings and shall be able to accommodate up to 2.5 mm sq. solid or stranded wire including spare points. The electrical ratings shall be 15 Amps/300 Volts and dielectric withstand shall be 3000 Volts.

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14.0 DC ANALOGUE INPUT SUBSYSTEM

14.1 The DC Analog Input Subsystem shall consist of logic and termination modules to process and monitor plant parameters derived from plant transducers.

14.2 The DC Analog Input Subsystem must support differential inputs to provide maximum noise immunity.

14.3 The RTU shall contain analogue-to-digital conversion equipment necessary to meet the analogue conversion rates identified by these specifications.

14.4 Analogue inputs may be positive or negative unipolar, bipolar, with or without offset and are converted to a digitized value, with polarity bit. A/D conversion shall be with a minimum resolution of 12 bits, plus the sign bit for bipolar conversion giving an overall range of negative count to positive count.

14.5 Over-range detection shall be provided.

14.6 TNB's standard practice is to use transducers with 4 to 20mA DC output to measure power system parameters. As such, as a default, the RTU's DC Analogue Input Sub-System must support 4 to 20mA dc inputs and shall come factory-fitted with precision scaling resistors.

14.7 The RTU shall also be able to be support to accept analogue inputs in the form of ±10mA, ±5mA, ±1mA, ±20mA, ±5 Volts, ±1 Volts and ±10 Volts.

14.8 It shall be possible to remove or replace scaling resistors at site without any re-soldering.

14.9 Programmable gain factor shall be employed to enable a range of current inputs to be used.

14.10 The overall accuracy of the DC Analogue Input Sub-system, from input terminal to digital value shall be better than ±0.1% for current inputs and ±0.05% for voltage inputs, of full scale within the specified operating range of the equipment, with less than +/-0.005% per deg. C temperature error. In the definition of accuracy, "full scale" is the measurement span, which is the algebraic difference of maximum positive and negative reading.

14.11 The DC Analogue Input Sub-system shall exhibit common mode noise rejection characteristics of at least 90dB from 0 to 50Hz (100 V p-p).

14.12 The normal (differential) mode rejection shall be at least 60 dB at 50 Hz (15V p-p).

14.13 Each AI module shall be equipped with an integrity check and auto calibration feature with at least two precise reference inputs at both ends of the full-scale analogue input. The long-term stability of the references shall be better than 0.02% of full scale over the specified range of the operating environment of the equipment. The references shall be used for auto calibration of the input

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and analogue-to-digital circuitry.

14.14 All analog inputs shall be protected against ground loops and power surges.

14.15 For current inputs, the input impedance shall be such that the voltage across the input terminals does not exceed 5 volts with full-scale input current (Ref. IEC 60870-3 Table 12), and no damage shall occur for sustained 100% overcurrent.

14.16 For voltage inputs, the input impedance shall not be less than 200 kΩ per volt (Ref. IEC 60870-3 Table 12).

14.17 It shall be possible to replace the logic modules without breaking any analog input current loops in the RTU.

14.18 Additionally, the Analog Input Subsystem shall be designed to facilitate the replacement of modules without necessitating the re-termination and re-testing of field cabling.

14.19 It shall be possible to configure the analog deadband (jitter/threshold) for each analog input point. The purpose of the analog deadband is to prevent noisy input signals from continuously reporting analogue fluctuations within the RTU and to the Master Station. The RTU shall only report new analog values that exceed the set deadband.

14.20 At the RTU, the analog deadband shall be configurable in the range of 0 to at least 5% of full scale.

14.21 The RTU shall be able to accept analog deadband parameter that is downloaded by the Master Station. (Refer IEC 60870-5-101 Parameter Download).

14.22 To eliminate noise fluctuations around the zero value, it shall be possible to configure the Force To Zero value for each analog input point such that the AI Sub-system shall report a zero value for any readings within zero and the Force to Zero value.

14.23 The Force To Zero parameter shall be configurable from 0 to at least 5% of full scale.

14.24 It shall be possible to configure a smoothing factor for each analog input to ensure that fast transient interference does not cause large fluctuations in the actual reported value that could result in incorrectly reporting limit transgressions.

The smoothing factor, F, shall be applied as shown:-

$$Value_{SMOOTH,\,NEW} = F*Value_{NEW} + (1-F)*Value_{SMOOTH,\,OLD}$$

$$= F(Value_{NEW} - Value_{SMOOTH,\,OLD}) + Value_{SMOOTH,\,OLD}$$

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14.25 The RTU shall be able to accept the smoothing factor parameter that is downloaded by the Master Station. (Refer IEC 60870-5-101 Parameter Download).

14.26 Using the RTU Maintenance Tools, it shall be possible to manually disable reporting of analogue input points.

14.27 Suitable terminal blocks with isolation and test points complete with cabling, shall be used for termination of field I/O cablings and shall be able to accommodate up to 2.5 mm sq. solid or stranded wire including spare points. The electrical ratings shall be 15 Amps/300 Volts and dielectric withstand shall be 3000 Volts.

15.0 CONTROL OUTPUTS SUBSYTEM

15.1 The RTU shall be used to operate various plant devices (TRIP/CLOSE circuit breakers, OPEN/CLOSE motorized switches, RAISE/LOWER Transformer Tap changers, RESET relays, etc.).

15.2 The RTU control output point shall be in the form of normally open relay contacts. Transistor output is not acceptable.

15.3 Secure Control Operation

15.3.1 For secure control, the RTU shall be designed to be fail-safe. The following minimum design criteria shall be incorporated in the control output logic:

- Only ONE control point on a control output module can be selected or executed at any given time.
- NO false output shall result during power up, power down or RTU reset.
- The control selection shall be automatically cancelled or overridden if, after receiving the `control selection' message, the `execute' command is not the next received message and/or it is not received within a predetermined configurable time-out period.
- NO false output shall result from a single point of failure.
- The verification message for a check-before-operate control sequence shall be obtained by re-encoding the verification message directly from the control point selection lines. A simple echoing of the received message is not acceptable.
- The RTU's control output shall be incorporate protection to guard against false output selection due to a single component failure, erroneous address or selection decoding.

15.3.2 For secure control of circuit breakers, switches and transformer tap changers, the RTU shall support select-before-operate control and the control outputs shall be configured as TRIP/CLOSE pairs.

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15.3.3 For TRIP/CLOSE points, each control output momentary control point shall comprise a paired control output consisting of a trip output and a corresponding close output.

15.3.4 Select-Before-Operate (SBO) Controls shall be used for secured operation, such as the Trip and Close operations of circuit breakers. Only one point can be controlled at a time. Each command requires that the following steps be performed in sequence:

- Device Selection: Upon receipt of the Master Station command message, the specified device is selected.
- Device Selection Validation: After the device is selected, the address is re-encoded and sent to the Master Station for validation.
- Control Execution: Upon receipt of the command execution message from the Master Station, the command is executed.
- Automatic Reset: If the command message is not received within a specified period, the device is de-selected.

15.4 For the automatic reset described above, the period shall be user-configurable and shall be able to accommodate typical communication delays.

15.5 The control output point's contact closure duration shall be user-configurable at the RTU.

15.6 Pulse duration for "Long Pulse" and "Short Pulse" (IEC 60870-5-101 Control Command) shall be user configurable.

15.7 Direct-Operate (Pulse Output) Controls are typically used for controlling devices and systems that do not require the control security. They can be configured as single or raise/lower pair output. Direct-Operate (Pulse Output) Controls shall be used for "step" functions such as precisely timed RAISE and LOWER commands to generator controllers. Multiple RAISE/LOWER points can be controlled simultaneously. On receipt of a command message from the Master Station, a timed pulse is sent to a specified device. These controls shall be configurable for latching or pulsed operation.

15.8 It is preferred that the RTU supports the Direct-Operate (Pulse Output) Control functionality using the same module as the Secure Control. The control mode, either Secure or Direct-Operate, shall be configurable using software or hardware methods. The Direct-Operate Controls can, however, be provided as a separate module than the Secure Control module.

15.9 In multiple card setup, each Control Card shall be equipped with a dummy breaker (latching relay) as a test indication for control function. One pair digital output in each Control Card shall be used for the TRIP and CLOSE output from the master station. The status of the relay shall be acquired by the RTU as a digital input for transmission to the master station. These control points and indicators are included in the specified RTU point counts. The status of the dummy breaker shall also be indicated visually using indicators

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to allow visual determination of the state of the dummy breaker.

15.10 Each RTU shall be equipped with a Control Enable Switch (CES) to connect or disconnect the power to all digital output relays, i.e. to enable or inhibit all digital output points without shutting off the RTU.

15.11 An auxiliary contact shall be provided on the CES to be wired to one digital input point to provide remote indication of the switch(s) status.

15.12 An audible alarm shall be designed in the CES circuit to indicate if the CES were left in the Disabled state when the RTU panel door is shut.

15.13 Suitable terminal blocks with isolation and test points complete with cabling, shall be used for termination of field I/O cablings and shall be able to accommodate up to 2.5 mm sq. solid or stranded wire including spare points. The electrical ratings shall be 15 Amps/300 Volts and dielectric withstand shall be 3000 Volts.

16.0 ANALOG OUTPUT SUBSYSTEM

The RTU shall support the Analog Output function, unless stated otherwise at the tendering stage.

The RTU shall provide the capability to output analog values (voltage or current) proportional to digital data received from the Master Station. Analog outputs are used for such functions as set points for proportional controllers, and data outputs to chart recorders.

17.0 SUBMASTER CAPABILITY

The RTU shall support the Submaster function, unless stated otherwise at the tendering stage.

The protocols for communication with Satellite RTUs or IEDs shall be specified at the project stage.

17.1 The RTU shall have capability to operate as a host/submaster to several remote Satellite RTUs or IEDs, with no additional changes to the firmware and software. It shall be possible to map the data from the Satellite RTUs or IEDs for onward transmission to any of the control centers that are connected to the Submaster RTU.

17.2 The software and the hardware in the Submaster RTU must be able to be expanded to allow the RTU to poll at least 16 Slave RTUs. The Satellite RTUs may comprise of RTUs installed in any of TNB's substations that may include PMUs, PPUs, SSUs and PEs or any IEDs in the same substation.

17.3 The RTU shall scan the databases of its Satellite RTUs or IEDs and also

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support local I/O. At any time, the master station can retrieve the information from the Submaster RTU.

17.4 The Submaster RTU shall have the capability of polling point-to-point and multidrop Satellite RTUs or IEDs simultaneously via separate communication ports.

17.5 The RTU shall be able to support communication with its satellite RTUs or IEDs using duplicated channels. The RTU shall initiate communication to the satellite RTU using the alternate channel when there is a failure on the main channel.

17.6 From the RTU Diagnostic tool, it shall be possible to monitor the data of the Satellite RTUs or IEDs as well as execute control commands from the Submaster RTU.

17.7 The Submaster RTU shall monitor the communication status to its Satellite RTUs and IEDs. When the Satellite RTUs or IEDs fail to respond, the Submaster RTU shall raise an alarm to the Control Center. The alarm shall be sufficient to allow the operator at the Control Center to identify the Satellite RTU/ RTUs or IEDs that had failed.

18.0 LOGIC OPERATIONS REQUIREMENTS

The RTU shall support the Logic Operation function, unless stated otherwise at the tendering stage.

18.1 The RTU shall be capable of being programmed to perform logical, mathematical, and trigonometric operations within the program on accumulator and input values, and store results in memory. The memory locations must be accessible to all programmed algorithms.

18.2 A programming tool that is compliant with the Open Standard PLC programming languages to IEC 61131-3 standard shall be used to develop logic programs for the RTU. Proof of compliance is mandatory.

18.3 The programming tool shall support the IEC 61131-3 programming model that includes Configuration, Resource, Task, Programs and Functions, Functional Block Diagrams, Ladder Diagram and Structured Text.

18.4 The functions supported shall be adequate to allow the user to program standard substation automation functions.

18.5 The programming tool shall provide the capability of developing new functions blocks and libraries in addition to the Standard Function Blocks recommended by the IEC-61131 Standard.

18.6 All I/O points within the RTU, including virtual I/O data and I/O data from satellite RTUs or IEDs shall be made accessible to the programming tool to be used in the programs

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18.7 The RTU shall be capable of storing and executing multiple algorithms which may be event-driven or periodic at selectable periods and priorities.

18.8 Algorithms residing in the RTU shall have access to the internal RTU clock for controlling time sensitive operations.

18.9 Vendors shall declare the limitations of the RTU such that normal RTU processes such as data processing and control are not affected by the extra processing required to run the logic programs.

18.10 The memory area in which the programmable logic algorithm executes shall be protected. This feature shall protect the memory area of the RTU to prevent any malfunction or system deadlocks and synchronization problems.

18.11 The programming tool shall have the facility to diagnose, de-bug and simulate programming logic programs.

18.12 Provision for training and the supply of the Programming software will be as per Tender basis.

19.0 RTU MAINTENANCE AND CONFIGURATION TOOLS REQUIREMENTS

19.1 The RTU shall be equipped with at least one (1) Maintenance and Configuration port.

19.2 The RTU shall be implemented a built-in web-based maintenance and configuration tools to be accessed via a secure web browser.

19.3 It shall be possible to connect the tools to and from any of the Ethernet port of the RTU, and use it to perform the RTU management, configuration, monitor stored data, downloading and uploading of all RTU files and SOEs and all RTU inputs, to exercise RTU outputs and to diagnose/troubleshoot the RTU, locally or remotely.

19.4 The tools shall utilize secure connection with standard access control procedures complying with Para 5.0 above.

19.5 The tools shall be used to monitor communications between the Master Station and the RTU, and the Submasters.

19.6 During online monitoring, the normal operation of the RTU to the Master Station shall not be disrupted in any way. Initiation of control shall not be possible in the online monitoring mode.

19.7 For Distributed I/O modules, it shall be possible to use the tools to monitor the inputs, control outputs, run local diagnostics, display the date and time of the real-time clock of the I/O module and monitor communications with the Master Processor module.

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19.8 For networked RTU, it shall be possible to connect the tools at any node and carry out all the test and monitoring functions as if the test equipment is connected to the main node.

19.9 Any RTU software tools supporting the maintenance and configuration of the RTU shall be fully common-licensed to TNB with unlimited device installation schemes.

19.10 All RTU modules shall have an ISO/IEC 15416 barcode printed on for RTU Maintenance Asset Tagging purposes. The information contain in the barcode will be determined in the implementation stage.

20.0 RTU CONFIGURATION SYSTEM REQUIREMENTS

20.1 A RTU Configuration Software, if not integrated with the web-based maintenance and configuration tools above, shall be supplied to allow TNB to build, configure and program the RTU, using a Windows based machine.

20.2 The software shall utilize secure standard access control procedures complying to Para 5.0 above.

20.3 All configuration, data and application software shall be stored with revision control information.

20.4 The Configuration software shall allow the compiled RTU configuration files to be downloaded and uploaded to the RTU via the maintenance tools.

20.5 The vendor shall provide an itemized test equipment and tools to support and run the configuration system, and the maintenance and configuration tools, with all the necessary license required.

20.6 The RTU software tools supporting the maintenance and configuration of the RTU shall be fully common-licensed to TNB with unlimited device installation schemes, inclusive of future version upgrades.

21.0 RTU PANEL

The type of the RTU panel required shall be specified during project stage on per project basis. The RTU panel shall comply with the general requirement as stated below:

21.1 General Requirement

21.1.1 Locally manufactured panels shall be used for all RTU panels.

21.1.2 The design and the specification for the panel shall be based on the RTU manufacturer standard panel. The local manufacturer shall be certified to the requirements of ISO9000 standard.

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21.1.3 The design and finish of the panel shall not invalidate any relevant type test requirements specified for this project.

21.1.4 The design of panels shall conform to IEC 60297 equipment practice unless otherwise approved.

21.1.5 The panels shall be of fabricated steel construction providing a good combination of high load carrying capacity and rigidity.

21.1.6 The panels shall be of modular construction with minimum weld joints. The modular construction should enable easy disassembly and reassembly of the panel to facilitate transportation and erection.

21.1.7 The panel shall provide optimum protection against corrosion.

21.1.8 The surface finish shall be dip-coat primed and powder coated. The interior of panel shall be finished with a matte white surface. The exterior finish shall be dip-coat primed and powder coated in textured RAL 7032.

21.1.9 The panels shall feature sheet steel construction with removable side panels. The material used shall be sheet metal of minimum thickness of 2 mm. for the door, side and rear panels. For the top and bottom plate, the minimum thickness of the sheet metal shall not be less than 3 mm.

21.1.10 Each panel shall have removable hinged, lockable front doors for access.

21.1.11 A master key shall be provided for all panels.

21.1.12 Front door shall be capable of being opened as wide as necessary to provide unimpeded access into the RTU enclosure. Range of 170 degrees to 180 degrees is acceptable.

21.1.13 All components, wiring and terminal blocks shall be accessible from the front of the panel. No rear access shall be required.

21.1.14 For panels with a swing frame:- All cables, wiring, etc. associated with the swing frame shall be secured, so as not to kink, bend or break. The swing frame shall be capable of being opened fully to allow unobstructed working access into the rear of the cabinet. The swing frame shall be furnished with a locking device to hold it in the open position and a latch for the closed position. The frame shall be of adequate strength to support mounted components.

21.1.15 The design and construction of the panel shall cater for both top and bottom cable entry with detachable gland-plates.

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21.1.16 Each panel shall be equipped with removable, interchangeable gland plates for easier cable insertion and handling, including pre-molded holes.

21.1.17 Internal illuminations and 2 units of 3-pin plug socket 13A shall be provided. Fluorescent lighting shall not be used.

21.1.18 Heat dissipation of panel mounted equipment shall be kept as low as possible and shall employ natural cooling only. As such, the RTU front and rear panels shall be fitted with ventilation louvers located at the top and bottom of the panels to ensure that the temperature rise within the panel does not exceed 15 deg C, with equipment operational. All ventilation openings shall have a metal wire mesh or other adequate protection, in order to provide protection against vermin and insects.

21.1.19 The panel shall be equipped with suitably rated miniature circuit breakers for AC distribution, RTU DC power distributions and modems.

21.1.20 Each panel shall be equipped with an earth bar with dimension at least 25mm (wide) x 3mm (thickness), mounted at the bottom of the panel, for electrical and equipment grounding purposes. Each grounding bar shall be equipped with pre-drilled holes and equipped with appropriate bolt and nuts for termination via a grounding cable to the general earthing system of the substation.

21.1.21 Electrostatic Discharge (E.S.D) bonding point complete with earth strap shall be fitted on the front of enclosures to provide an obvious and readily visible earth bond point. The bonding point shall be wired directly to ground via an eyelet on the rear and shall be electrically isolated from the enclosure by means of an insulator.

21.1.22 Alarm lamps for RTU common alarm indication and DC Supply indication shall be fitted on the front of enclosures.

21.1.23 Cable channels and cable guide rings shall available to install the cables on the mounting panels and the side walls.

21.1.24 A metal pocket suitable for storing all site-specific documentation and drawings shall be provided within the enclosure, on the front door. The size of the pocket shall be 300mm height, 420mm width and 20mm depth.

21.1.25 A stainless steel nameplate shall be furnished for each RTU panel and shall be attached to the panel with stainless steel panhead screws. Nameplate identifiers for each RTU will be provided after the award of contract. The lettering shall be permanently engraved onto the nameplate.

21.1.26 A retractable shelf or equivalent tray shall be provided in the panel to place RTU maintenance tools and at a proper height.

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21.2 Floor Standing RTU Panel

21.2.1 The RTU panel shall be floor standing and shall permit anchoring to the floor.

21.2.2 The size of the panel, inclusive of plinth, shall be 2300mm in height, 800mm in width and 800mm in depth.

21.2.3 The panel shall conform to minimum IP 51 protection from dust and splashing water. A type test shall be carried out to show compliance to the IP51 protection. The type test certificates/reports shall be made available during Product Inspection.

21.2.4 All floor standing panel shall be fitted with 100mm plinths for mounting purposes and cable entry.

21.2.5 Two modem shelf units in 2U height shall be provided. The location shall be ergonomic and accessible.

21.3 Wall-Mounted RTU Panel (standard size)

21.3.1 For substations with space constraints, wall-mounted RTU panel shall be used.

21.3.2 The size of the panel shall be 700mm-900mm in height, 600mm-800mm in width and 300mm-400mm in depth. The size of panel will be finalized during engineering design approval stage.

21.3.3 The panel shall conform to minimum IP 51 protection from dust and splashing water. A type test shall be carried out to show compliance to the IP51 protection.

21.3.4 One modem shelf unit in 2U height shall be provided. The location shall be ergonomic and accessible.

21.4 Outdoor RTU Panel

21.4.1 The RTU panel either floor-standing or wall-mounted shall be specified during project stage.

21.4.2 The panel shall conform to minimum IP 54 protection from dust and splashing water. A type test shall be carried out to show compliance to the IP54 protection. The type test certificates/reports shall be made available during Product Inspection.

21.4.3 The panel shall be made of galvanized mild steel construction with removable side panels. The material used shall be with nominal thickness of not less than 3mm. Other hardware such as hinges, supports and screws shall be made from stainless steel.

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21.4.4 The padlock facilities shall be provided with hole for shackles not less than 12 mm diameter to accommodate typical padlock.

21.4.5 An anti-theft plate that covers the padlocking facilities shall be provided with minimum thickness 3 mm which is riveted securely to the door. The plate shall resemble cubic form with opening provided only at the bottom with sufficient clearance to allow padlock to be inserted/taken out. The dimension of the plate shall be able to cover the entire inserted padlock of length 100 mm and width 70 mm.

21.4.6 The panel shall provide space modem shelf unit in 2U height and rail for TNB-issued modem's and its accessories. A pre-molded hole for antenna shall be provided complete with IP54 gland.

21.4.7 There shall be adequate space for cabling and termination of cables inside the panel. All wirings inside the panel shall be supported by cable raceway.

21.4.8 Roof facility shall be provided over the door access to the Secondary Enclosure Panel with specifications as follow:

(i) Does not affect design integrity of the enclosure such as IP54
(ii) Retractable by sliding so as can be used as an option and not permanently set up
(iii) Maintenance-free material which is able to withstand outdoor weather
(iv) Easy and fast to set up and does not require additional/specialised tools
(v) The proposed dimension of the roof when fully extended is as follow:
- Length ≥ 1.0 m
- Width shall match width of enclosure
- Thickness of roofing sheet ≥ 1.0 mm

21.4.9 The size for floor standing panel shall be 2300mm in height, 800mm in width and 800mm in depth, inclusive of plinth.

21.4.10 The size of the panel shall be 700mm-900mm in height, 600mm-800mm in width and 300mm-400mm in depth. The size of panel will be finalized during engineering design approval stage.

21.4.11 As and when required, the panel shall be securely installed on a concrete-based plinths with a height of at least 500mm from the ground level and shall ready for cable entry from Galvanized-iron (GI) pipes.

21.4.12 The panel shall be in TNB's corporate colour and painted with non-stick paint of brands and types from TNB approved list.

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22.0 RTU CABLING

The RTU cabling shall comply with the requirement as stated in latest Technical Specification for Telecontrol System Cables & Cabling. The type and method of the RTU cabling shall be specified during project stage on per project basis.

23.0 DRAWINGS AND DOCUMENTATIONS

Comprehensive RTU documentation shall be provided. The documentations shall include all information and instruction needed to install, configure, troubleshoot and repair of the RTU. RTU documentation shall be available in hardcopies and softcopies.

24.0 RTU FACTORY ACCEPTANCE TEST (FAT)

The purpose of the FAT is to qualify the RTU and all of it subsystems and all other related deliverables that are required according to the specifications of the Project.

The document 'Dasar Jaminan Kualiti Produk TNB' shall be the reference document for this purpose.

The scope of the FAT shall include the followings:

- Visual inspection to confirm that construction and sizing requirements have been met.
- Demonstrate compliance to specific requirements of the Project not covered by the TNB RTU Technical Specifications.
- Demonstrate that the RTUs are assembled and configured according to the requirements of the Project.
- Demonstrate the functional functions for RTU modules points including spare points via a comprehensive test procedures.
- Verification of all communication ports and RTU parameters.

25.0 RTU EMULATOR/SIMULATOR

As an optional item, TNB requires tenderer to offer a RTU software emulator/simulator running on Microsoft Windows environment or as a web-based application, as a diagnostic, investigative, training tools.

The emulator shall behave as a physical complete RTU modules capable to be configured using relevant RTU firmwares and softwares and running IO subsystem with all the required protocols via virtual communication ports.

----------End Of Specification----------

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